

Hands-On: Controlled vs Uncontrolled Components in React

In this lab, you will create two simple React components, one that is a controlled component and one that is an uncontrolled component.

Step 1: Create a new controlled component

Create a new file called `ControlledForm.js` and add the following code:

```
import React, { Component } from 'react';

class ControlledForm extends Component {
  state = {
    inputValue: "",
  };

  handleInputChange = e => {
    this.setState({ inputValue: e.target.value });
  };

  handleSubmit = e => {
    e.preventDefault();
    alert(`You entered: ${this.state.inputValue}`);
  };

  render() {
    const { inputValue } = this.state;

    return (
      <div>
        <h1>Controlled Component</h1>
        <form onSubmit={this.handleSubmit}>
          <label>
            Enter some text:
            <input type="text" value={inputValue} onChange={this.handleChange} />
          </label>
          <button type="submit">Submit</button>
        </form>
      </div>
    );
  }
}

export default ControlledForm;
```

This component is a controlled component, meaning that the input value is controlled by the component's state. The `handleInputChange` method updates the component's state with the input value, and the `handleSubmit` method alerts the user with the input value when the form is submitted.

Step 2: Create a new uncontrolled component

Create a new file called ``UncontrolledForm.js`` and add the following code:

```
import React, { Component } from 'react';

class UncontrolledForm extends Component {
  handleSubmit = e => {
    e.preventDefault();
    const inputValue = this.input.value;
    alert(`You entered: ${inputValue}`);
  };

  render() {
    return (
      <div>
        <h1>Uncontrolled Component</h1>
        <form onSubmit={this.handleSubmit}>
          <label>
            Enter some text:
            <input type="text" ref={input => (this.input = input)} />
          </label>
          <button type="submit">Submit</button>
        </form>
      </div>
    );
  }
}

export default UncontrolledForm;
```

This component is an uncontrolled component, meaning that the input value is not controlled by the component's state. The input field has a ``ref`` attribute that is used to access the input value in the ``handleSubmit`` method.

Step 3: Render the components

Create a new file called ``App.js`` and add the following code:

```
import React from 'react';
import ControlledForm from './ControlledForm';
import UncontrolledForm from './UncontrolledForm';

function App() {
  return (
    <div className="App">
      <ControlledForm />
      <UncontrolledForm />
    </div>
  );
}

export default App;
```

This component simply renders the ``ControlledForm`` and ``UncontrolledForm`` components.

Step 4: Run the app

Run the app using `npm start` and test the `ControlledForm` and `UncontrolledForm` components by entering some text into the input field and submitting the form.

Congratulations! You have successfully created a controlled and an uncontrolled component in a React app.